Factorising and Expanding Brackets

Four tiers: work through Fluency first, then Problem Solving and Reasoning in either order, then try Enrichment as a challenge.

Fluency

Expand or factorise each expression. One bracket, no application.

Expand

(a) $3(x+4)$

(b) $5(2x-3)$

(c) $-2(x+6)$

(d) $4(3x-5)$

(e) $-(x-7)$

(f) $x(x+3)$

(g) $2x(x-5)$

(h) $-3x(2x+1)$

Factorise

(a) $4x+12$

(b) $6x-15$

(c) x^2+5x

(d) $3x^2-9x$

(e) $10x+25$

(f) $2x^2+8x$

(g) $9x-6$

(h) $9x^2-6x$

Problem Solving

These need more than one step: multiple brackets, variable common factors, or an equation to solve.

Expand and simplify

(a) $3(x+2)+2(x-5)$

(b) $4(2x-1)-3(x+2)$

(c) $(x+3)(x+4)$

(d) $(2x-1)(x+5)$

(e) $(x-6)(x-2)$

Factorise fully

Find the highest common factor (HCF) of every term — it can include more than one letter, and a power of a letter.

(a) $18x^5y^2+12x^7$

(b) $8a^3b^2-20a^2b^4$

(c) $15x^3y-9x^2y^2+6xy^3$

(d) $14p^4q^3+21p^2q^5$

(e) $24m^6n^2-16m^4n^5+8m^3n^3$

Solve

Expand the brackets first, then solve for x.

(a) $3(x+5)=27$

(b) $2(3x-4)=5x+1$

(c) $5(x-2)-3(x+1)=7$

A step further

(a) A rectangle has length $(x+5)$ and width $(x-2)$. Write an expression for its area, expanded and simplified.

(b) The length of a rectangle is $2x-3$ and its width is x. The area is $2x^2-3x$. Factorise this area expression to check it matches the length and width given.

Reasoning

No variables here — use expanding or factorising as a mental-maths trick to work these out without *a calculator. Show the expanded/factorised form you used.*

(a) Write 99 as $(100-1)$ and 101 as $(100+1)$. Use the difference of two squares to calculate 99×101.

(b) Use factorising to quickly calculate $37 \times 8 + 37 \times 2$.

(c) Write 52 as $(50+2)$ and 48 as $(50-2)$. Use the difference of two squares to calculate 52×48.

(d) Write 203 as $(200+3)$. Use expanding to calculate 6×203.

(e) A school has 29 classes with 31 students in each. Write 29 as $(30-1)$ and 31 as $(30+1)$, then use the difference of two squares to find the total number of students.

(f) Write 999 as $(1000-1)$. Use expanding to calculate 999^2.

Enrichment

Factorising Quadratics

A quadratic x^2+bx+c (where x^2 has no coefficient) factorises as $(x+p)(x+q)$, where p and q are two numbers that multiply *to give c and* add *to give b. If the quadratic has a coefficient in front of the x^2, like ax^2+bx+c, find two numbers that multiply to give $a \times c$ and add to give b, use them to split the middle term into two terms, then factorise by grouping. A difference of two squares, a^2-b^2, always factorises as $(a-b)(a+b)$.*

Worked example (no coefficient): Factorise x^2+5x+6. Two numbers that multiply to 6 and add to 5 are 2 and 3, so $x^2+5x+6=(x+2)(x+3)$.

Worked example (with a coefficient): Factorise $2x^2+7x+3$. Here $a \times c = 2 \times 3 = 6$, and we need two numbers that multiply to 6 and add to 7: these are 1 and 6. Split the middle term and group:

$$2x^2+7x+3 = 2x^2+x+6x+3 = x(2x+1)+3(2x+1) = (2x+1)(x+3).$$

(a) $x^2+7x+10$

(b) $x^2-3x-18$

(c) x^2-16

(d) $4x^2 - 9$

(e) $2x^2 + x - 6$

Inequalities

Inequalities behave like equations, except multiplying or dividing by a negative number reverses the inequality sign.

(a) Solve: $3(x + 2) > 15$

(b) Solve: $2(x - 5) \leq 4(x + 1)$

(c) Solve: $-3(x + 1) < 2(x - 4)$

(d) Solve the double inequality: $2 < 3(x - 1) \leq 11$

(e) A taxi charges a \$4 call-out fee plus \$3 per kilometre travelled. Write an inequality for the total cost of a trip of k kilometres, then solve it to find how far you could travel and keep the cost under \$40.